



Sec: 5C-60

D-Block elements

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SINGLE ANSWER TYPE

- Which of the following has highest ionic radii?
A) Cr^{+3} B) Mn^{+3} C) Fe^{+3} D) Co^{+3}
- Which of the following element has maximum density?
A) Hg B) Au C) Os D) Pb
- The highest oxidation state achieved by a transition metal is given by
A) ns electrons B) $(n - 1)d$ electrons
C) $(n + 1)d$ electrons D) $ns + (n - 1)d$ electrons
- Which one of the following characteristics of transition metals is associated with their catalytic activity?
A) Paramagnetic behaviour B) colour of hydrated ions
C) Variable oxidation states D) high enthalpy of atomisation
- In the dichromate dianion
A) 4 Cr — O bonds are equivalent B) 6 Cr — O bonds are equivalent
C) All Cr — O bonds are equivalent D) All Cr — O bonds are non – equivalent
- What is the general electronic configuration for 2nd row transition series
A) $[\text{Ne}]3d^{1-10}, 4s^2$ B) $[\text{Ar}]3d^{1-10}, 4s^{1-2}$ C) $[\text{Kr}]4d^{1-10}, 5s^{1-2}$ D) $[\text{Xe}]5d^{1-10}, 5s^{1-2}$
- Which ion has maximum magnetic moment
A) V^{+3} B) Mn^{+3} C) Fe^{+3} D) Cu^{+2}
- Which of the following statement is not true about *Mohr's salt*?
A) It decolourises KMnO_4
B) It is a primary standard
C) It is a double salt
D) Oxidation state of iron is +3 in this salt
- In the first transition series, the highest B.P. and M.P. is of
A) Cr B) V C) Ni D) Fe
- The placement of Zn, Cd and Hg along with 'd' block elements is not proper because
A) Their 'd' orbitals are completely filled
B) Their 'd' orbitals are empty
C) They do not form complex compounds
D) They do not form coloured compounds
- Which of the following pair of ions may exhibit same colour
A) Cr^{3+} and Fe^{2+} B) Ti^{3+} and V^{2+} C) Fe^{3+} and Mn^{2+} D) Cu^{+} and Ni^{2+}
- When an acidified solution of ferrous ammonium sulphate is treated with potassium permanganate solution, the ion which is oxidised is
A) MnO_4 B) NH_4^+ C) Fe^{++} D) SO_4^{2-}
- CrO_4^{2-} (yellow) changes to $\text{Cr}_2\text{O}_7^{2-}$ (orange) in $\text{pH} = x$ and vice-versa in $\text{pH} = y$. Hence, x and y are:

A) 6,8

B) 6,5

C) 8,6

D) 7,7

14. CrO_3 dissolves in aqueous NaOH to give:
 A) $\text{Cr}_2\text{O}_7^{2-}$ B) CrO_4^{2-} C) $\text{Cr}(\text{OH})_3$ D) $\text{Cr}(\text{OH})_2$
15. The product of I^- with MnO_4^- in alkaline medium is:
 A) I_2 B) IO^- C) IO_3^- D) IO_4^-
16. On heating KMnO_4 , one among the following is not formed:
 A) K_2MnO_4 B) O_2 C) MnO_2 D) MnO
17. Which represents correct comparison of the stability of oxometallic anions?
 A) $\text{MnO}_4^- < \text{Mn}^{2+}$ B) $\text{Cr}^{2+} < \text{Cr}^{3+}$ C) $\text{CrO}_4^{2-} < \text{Cr}^{3+}$ D) All of these
18. Potassium dichromate in alkaline solution, with 30% H_2O_2 produces :
 A) K_3CrO_8 B) $\text{CrO}(\text{O}_2)_2$ C) CrO_3 D) K_2CrO_4
19. KMnO_4 on treatment with conc. H_2SO_4 forms a compound (X) which decomposes explosively on heating forming (Y). The (X) and (Y) are respectively :
 A) Mn_2O_7 , MnO_2 B) Mn_2O_7 , Mn_2O_3
 C) MnSO_4 , Mn_2O_3 D) Mn_2O_3 , MnO_2
20. Which of the following is true for the species having $3d^4$ configuration?
 A) Cr^{2+} is reducing in nature B) Mn^{3+} is oxidising in nature.
 C) Both (A) and (B) D) Need electrode potential data
21. One of the products formed due to the reaction between KMnO_4 and HCl is
 A) Red liquid, CrO_2Cl_2 B) black powder, MnO_2
 C) Greenish yellow gas, Cl_2 D) Colourless liquid, HClO_4
- MULTIPLE CORRECT OPTION TYPE**
22. Transition elements with $4s^1$ configuration are:
 A) Cr B) Zn C) Ni D) Cu
23. Which of the following statements is/are not true in regard to transition elements?
 A) They readily form complex compounds
 B) They show variable valency
 C) All their ions are colourless
 D) Their ions contain partially filled d -electron levels
24. Which of the following statements are correct?
 A) An acidified solution of $\text{K}_2\text{Cr}_2\text{O}_7$ liberates iodine from iodides
 B) In acidic solution dichromate ions are converted to chromate ions
 C) Ammonium dichromate on heating undergoes exothermic decomposition to give Cr_2O_3
 D) Potassium dichromate is used as a titrant for Fe^{2+} ions
25. Which of the following oxides are not acidic in nature
 A) Mn_2O_3 B) Mn_3O_4 C) MnO_2 D) Mn_2O_7
26. When CO_2 is passed into aqueous:
 A) Na_2CrO_4 solution, its yellow colour changes to orange
 B) K_2MnO_4 solution, its disproportionates to KMnO_4 and MnO_2
 C) $\text{Na}_2\text{Cr}_2\text{O}_7$ solution, its orange colour changes to green
 D) KMnO_4 solution, its pink colour changes to green
27. Pyrolusite is MnO_2 used to prepare KMnO_4 . Steps are, $\text{MnO}_2 \xrightarrow{\text{I}} \text{MnO}_4^{2-} \xrightarrow{\text{II}} \text{MnO}_4^-$. Steps I and II are respectively:
 A) fuse with KOH /air and electrolytic oxidation
 B) fuse with KOH/KNO_3 and electrolytic oxidation

- C) fuse with conc. HNO_3 /air and electrolytic oxidation
 D) dissolve in H_2O and oxidation
28. Potassium manganate (K_2MnO_4) is formed when:
 A) cuprous oxide reacts with alkaline KMnO_4 solution
 B) manganese dioxide is fused with K_2CO_3 in presence of KNO_3
 C) formaldehyde reacts with potassium permanganate in presence of concentrated potassium hydroxide solution
 D) potassium permanganate is heated with potassium hydroxide
29. Select the correct statement(s) with respect to oxides and oxoanions of transition metals.
 A) Among oxides of chromium, CrO is basic, Cr_2O_3 is amphoteric and CrO_3 is acidic.
 B) No higher oxides of iron above Fe_2O_3 are found.
 C) Ti, V, Cr and Mn form oxides MO and their correct increasing order of acidic character is $\text{MnO} < \text{CrO} < \text{VO} < \text{TiO}$
 D) Vanadium (v) oxide does not react with acids but reacts with alkali only.
30. Potassium manganate (K_2MnO_4) is formed when :
 A) Potassium permanganate reacts with formaldehyde in acidic medium.
 B) Manganese dioxide is fused with potassium hydroxide in air.
 C) Manganese hydroxide reacts with K_2O_2 in alkaline medium.
 D) Potassium permanganate is heated at 750K.
31. Which of the following statements are correct?
 A) Interstitial compounds have high melting points, higher than those of pure metals.
 B) Permanganate titrations in presence of hydrochloric acid are unsatisfactory.
 C) KMnO_4 does not act as an oxidising agent in strong alkaline medium.
 D) KMnO_4 on heating in a current of H_2 gives MnO .
32. Which of the following statements are correct?
 A) Acidic dichromate solutions on treatment with H_2O_2 gives deep blue $\text{CrO}(\text{O}_2)_2$ in presence of ether layer
 B) A deep red liquid, CrO_2Cl_2 is formed by the reaction of chromium(III) oxide with HCl in presence of conc. H_2SO_4 .
 C) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating yields green chromium(III) oxide and nitrogen gas.
 D) $\text{K}_2\text{Cr}_2\text{O}_7$ on heating with charcoal produces K_2CO_3 .
33. Which of the following statements are correct?
 A) Mercurous ion exists as Hg^+
 B) Mercurous ion is diamagnetic and exists as dimer Hg_2^{2+}
 C) Mercurous ion is colourless
 D) There is a covalent bond between two Hg^+ ions in Hg_2^{2+}

KEY

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| 1. A | 2. C | 3. D | 4. C | 5. B | 6. C | 7. C |
| 8. D | 9. A | 10. A | 11. A | 12. C | 13. A | 14. B |
| 15. C | 16. C | 17. D | 18. A | 19. A | 20. C | 21. C |
| 22. A | 23. CD | 24. ACD | 25. ABC | 26. AB | 27. AB | 28. ABCD |
| 29. ABD | 30. BCD | 31. ABD | 32. ACD | 33. BCD | | |